

CSMVessel0000000008 V2.0

Project Noah's Aegis: The Dielectric Citadel and the Future of Maritime Resilience

Version 2.0 | June 2026

Changes V1.0 to V2.0

- Synthesis document: integrates all CSMVessel 001-007 findings into unified rebuild standard
- DNV GL EMP-2 classification (2025 draft): full compliance roadmap
- Modular retrofit: phased ship-by-ship implementation strategy for 93-ship fleet
- Carbon intensity alignment: BFRP hull + Elium(r) reduces CII (Carbon Intensity Indicator)
- Solar Cycle 25 peak: urgency elevated; estimated storm window 2025-2027

1. The Complete Aegis Vessel — V2.0 Standard

1.1 Full Material Bill of Construction (per new-build or full retrofit)

System	V1.0 (Conventional)	V2.0 (Aegis)	GIC Status
Hull structure	AH36 steel	BFRP/Elium(r) inserts + ZrO2 isolation	IMMUNE
Hull coating	Conventional antifouling	YInMn Blue QD (SRI=130)	IMMUNE + anti-fouling
Propulsion pods	Steel/Al nacelle	BFRP nacelle + Si3N4 bearings	IMMUNE
Shaft stern seal	Steel bearing	Si3N4 ceramic bearing	ISOLATED
Deck wiring	Copper cable	PMMA POF signal + DC-blocked power	IMMUNE (signal)
Emergency power	Diesel + battery	LiFePO4 in BFRP/MXene enclosure	IMMUNE
Navigation	GPS + copper nav	GPS + LoRa backup + optical fiber nav	HARDENED
Life safety (SOLAS)	Copper wired	Optical fire detect + ceramic davit motors	IMMUNE
Entertainment	Conductive systems	GaN amps + BFRP cases + PMMA fiber	IMMUNE
Cabins	Standard construction	Aegis Embark transfer + Stellar-Tint	IMMUNE

1.2 Full Aegis Vessel Economics V2.0

For a new-build 300m cruise ship (capital cost ~\$800M): **Aegis upgrade premium over conventional:** \$38M-\$52M (4.7-6.5% of ship cost)

Revenue protection value: - Average 7-day cruise revenue: \$7.2M (93% occupancy, 5,200 passengers)
- Carrington event strands passengers at sea: lost revenue + rescue cost estimate \$45M-\$95M - 10% probability (SC25 decade): Expected value = \$4.5M-\$9.5M first decade - Aegis cost per ship: \$45M / 30-year life = \$1.5M/year amortized - Break-even: \$1.5M/year cost vs \$4.5M-\$9.5M expected storm benefit = **3-6x positive ROI**

1.3 DNV GL EMP-2 Classification Roadmap

Step 1 (12 months): Certify BFRP hull panel equivalency per ClassNK; initiate type approval **Step 2 (18 months):** Submit MXene enclosure design for ABS/DNV type approval **Step 3 (24 months):** First ship

pilot retrofit; DNV GL survey and system inspection **Step 4 (36 months)**: EMP-2 class notation awarded for pilot vessel **Step 5 (4-7 years)**: Roll-out across fleet with proven certified system

2. Market Opportunity — Global Cruise Fleet

The global cruise fleet in 2026: - 322 ocean-going cruise ships (CLIA 2026 report) - 237 river cruise vessels

Total Aegis retrofit market (ocean cruise only): 322 ships x \$45M avg full Aegis = **\$14.49B total addressable market**

At 15% market penetration in 10 years: **\$2.17B CSM revenue** (retrofit + supply chain)

CSM is the only company with the complete Aegis-certified material ecosystem to address this market — BFRP, YInMn Blue, MXene, Si₃N₄, LoRa, PMMA POF, LiFePO₄, KNbO₃-BaTiO₃ ultrasonic motors — all from a single supply chain. Competitive moat is complete.

End of CSMVessel0000000008 V2.0 | Carrington Storm Motors